

Chapter 5

Continuous-Time Fourier Transform

5.1 Introduction

The Fourier series provides an extremely useful representation for periodic signals. Often, however, we need to deal with signals that are not periodic. A more general tool than the Fourier series is needed in this case. In this chapter, we will introduce a tool for representing arbitrary (i.e., possibly aperiodic) signals, known as the Fourier transform.

5.2 Development of the Continuous-Time Fourier Transform

As demonstrated earlier, the Fourier series is an extremely useful signal representation. Unfortunately, this signal representation can only be used for periodic signals, since a Fourier series is inherently periodic. Many signals, however, are not periodic. Therefore, one might wonder if we can somehow use the Fourier series to develop a representation for aperiodic signals. As it turns out, this is possible. In order to understand why, we must make the following key observation. An aperiodic signal can be viewed as a periodic signal with a period of infinity. By viewing an aperiodic signal as this limiting case of a periodic signal where the period is infinite, we can use the Fourier series to develop a more general signal representation that can be used in the aperiodic case. (In what follows, our development of the Fourier transform is not completely rigorous, as we assume that various integrals, summations, and limits converge. Such assumptions are not valid in all cases. Our development is mathematically sound, however, provided that the Fourier transform of the signal being considered exists.)

Suppose that we have an arbitrary signal $x(t)$ that is not necessarily periodic. Let us define the signal $x_T(t)$ as

$$x_T(t) = \begin{cases} x(t) & \text{for } -\frac{T}{2} \leq t < \frac{T}{2} \\ 0 & \text{otherwise.} \end{cases} \quad (5.1)$$

In other words, $x_T(t)$ is identical to $x(t)$ over the interval $-\frac{T}{2} \leq t < \frac{T}{2}$ and zero otherwise. Let us now repeat the portion of $x_T(t)$ for $-\frac{T}{2} \leq t < \frac{T}{2}$ to form a periodic signal $\tilde{x}(t)$ with period T . That is, we define $\tilde{x}(t)$ as

$$\tilde{x}(t) = x_T(t) \text{ for } -\frac{T}{2} \leq t < \frac{T}{2} \quad \text{and} \quad \tilde{x}(t) = \tilde{x}(t + T).$$

In Figures 5.1 and 5.2, we provide illustrative examples of the signals $x(t)$, $x_T(t)$, and $\tilde{x}(t)$.

Before proceeding further, we make two important observations that we will use later. First, from the definition of $x_T(t)$, we have

$$\lim_{T \rightarrow \infty} x_T(t) = x(t). \quad (5.2)$$

Second, from the definition of $x_T(t)$ and $\tilde{x}(t)$, we have

$$\lim_{T \rightarrow \infty} \tilde{x}(t) = x(t). \quad (5.3)$$

Now, let us consider the signal $\tilde{x}(t)$. Since $\tilde{x}(t)$ is periodic, we can represent it using a Fourier series as

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad (5.4)$$

where

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jk\omega_0 t} dt \quad (5.5)$$

and $\omega_0 = 2\pi/T$. Since $x_T(t) = \tilde{x}(t)$ for $-\frac{T}{2} \leq t < \frac{T}{2}$, we can rewrite (5.5) as

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-jk\omega_0 t} dt.$$

Furthermore, since $x_T(t) = 0$ for $t < -\frac{T}{2}$ and $t \geq \frac{T}{2}$, we can rewrite the preceding expression for a_k as

$$a_k = \frac{1}{T} \int_{-\infty}^{\infty} x_T(t) e^{-jk\omega_0 t} dt.$$

Substituting this expression for a_k into (5.4) and rearranging, we obtain the following Fourier series representation for $\tilde{x}(t)$:

$$\begin{aligned} \tilde{x}(t) &= \sum_{k=-\infty}^{\infty} \left[\frac{1}{T} \int_{-\infty}^{\infty} x_T(\tau) e^{-jk\omega_0 \tau} d\tau \right] e^{jk\omega_0 t} \\ &= \sum_{k=-\infty}^{\infty} \left[\frac{\omega_0}{2\pi} \int_{-\infty}^{\infty} x_T(\tau) e^{-jk\omega_0 \tau} d\tau \right] e^{jk\omega_0 t} \\ &= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_T(\tau) e^{-jk\omega_0 \tau} d\tau \right] e^{jk\omega_0 t} \omega_0. \end{aligned}$$

Substituting the above expression for $\tilde{x}(t)$ into (5.3), we obtain

$$x(t) = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_T(\tau) e^{-jk\omega_0 \tau} d\tau \right] e^{jk\omega_0 t} \omega_0. \quad (5.6)$$

Now, we must evaluate the above limit. As $T \rightarrow \infty$, we have that $\omega_0 \rightarrow 0$. Thus, in the limit above, $k\omega_0$ becomes a continuous variable which we denote as ω , ω_0 becomes the infinitesimal $d\omega$, and the summation becomes an integral. This is illustrated in Figure 5.3. Also, as $T \rightarrow \infty$, we have that $x_T(t) \rightarrow x(t)$. Combining these results, we can rewrite (5.6) to obtain

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega,$$

where

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt.$$

Thus, we have found a representation of the arbitrary signal $x(t)$ in terms of complex sinusoids at all frequencies. We call this the Fourier transform representation of the signal $x(t)$.

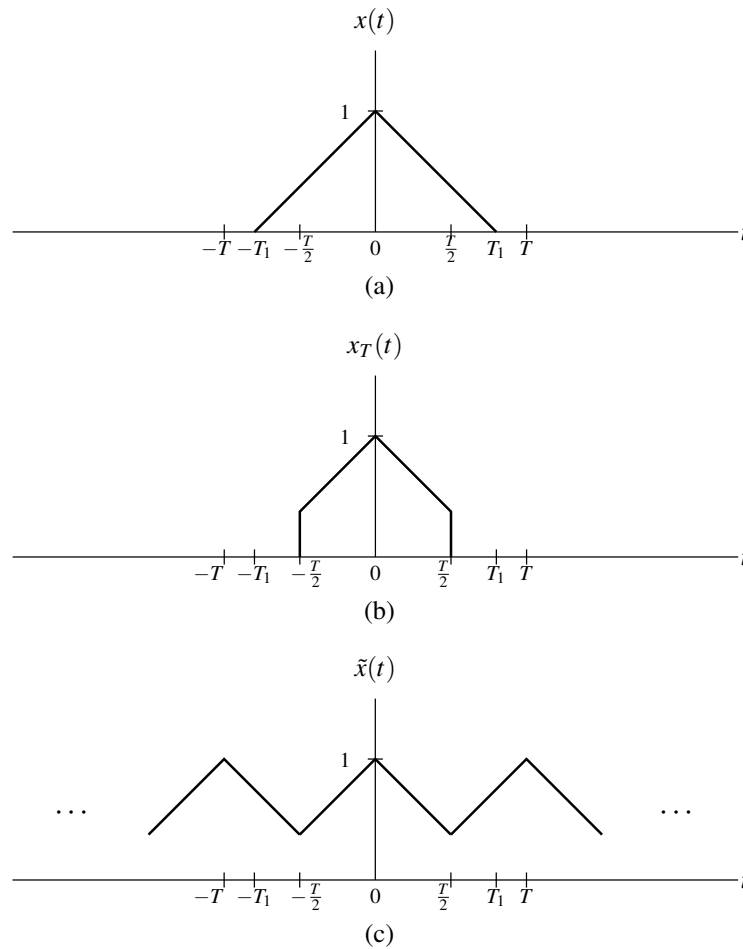


Figure 5.1: Example of signals used in derivation of Fourier transform representation (where $T_1 > \frac{T}{2}$).

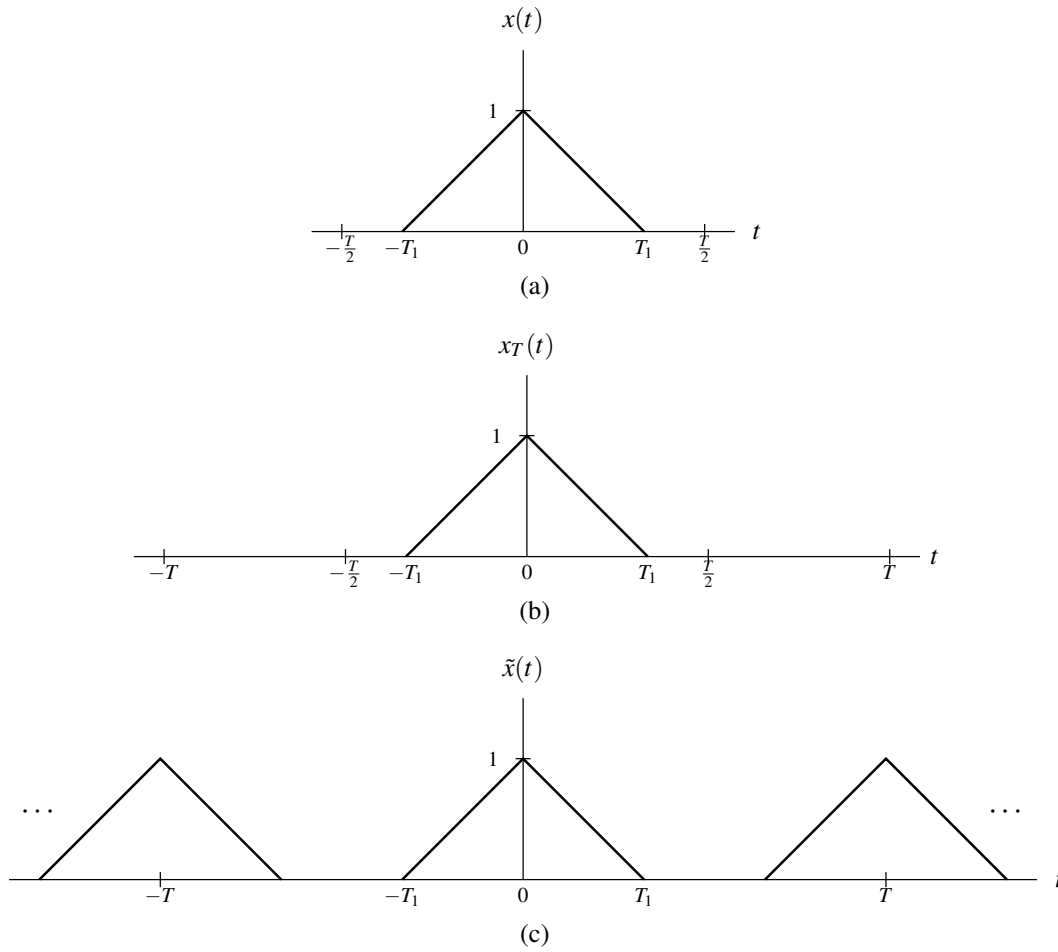


Figure 5.2: Example of signals used in derivation of Fourier transform representation (where $T_1 < \frac{T}{2}$).

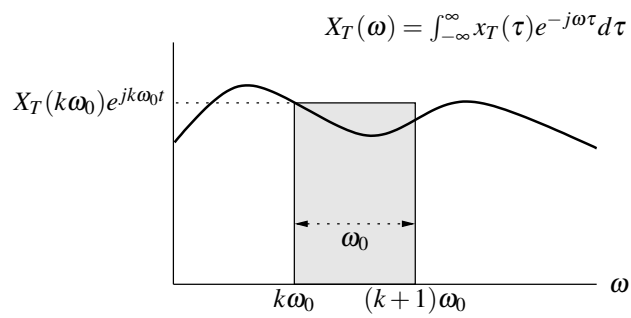


Figure 5.3: Integral obtained in the derivation of the Fourier transform representation.

5.3 Definition of the Continuous-Time Fourier Transform

In the previous section, we derived the Fourier transform representation of an arbitrary signal. This representation expresses a signal in terms of complex sinusoids at all frequencies. Given a signal $x(t)$, its Fourier transform representation is given by

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega, \quad (5.7a)$$

where

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt. \quad (5.7b)$$

We refer to (5.7b) as the **Fourier transform analysis equation** and (5.7a) as the **Fourier transform synthesis equation**.

The quantity $X(\omega)$ is called the Fourier transform of $x(t)$. That is, the Fourier transform of the signal $x(t)$, denoted as $\mathcal{F}\{x(t)\}$ or $X(\omega)$, is defined as

$$\mathcal{F}\{x(t)\} = X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt. \quad (5.8)$$

Similarly, the inverse Fourier transform of $X(\omega)$ is given by

$$\mathcal{F}^{-1}\{X(\omega)\} = x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega. \quad (5.9)$$

If a signal $x(t)$ has the Fourier transform $X(\omega)$ we often denote this as

$$x(t) \xleftrightarrow{\mathcal{F}} X(\omega).$$

As a matter of terminology, $x(t)$ and $X(\omega)$ are said to be a **Fourier transform pair**.

Example 5.1 (Fourier transform of the unit-impulse function). Find the Fourier transform $X(\omega)$ of the signal $x(t) = A\delta(t - t_0)$. Then, from this result, write the Fourier transform representation of $x(t)$.

Solution. From the definition of the Fourier transform, we can write

$$\begin{aligned} X(\omega) &= \mathcal{F}\{x(t)\} \\ &= \mathcal{F}\{A\delta(t - t_0)\} \\ &= \int_{-\infty}^{\infty} A\delta(t - t_0) e^{-j\omega t} dt \\ &= A \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt. \end{aligned}$$

Using the sifting property of the unit-impulse function, we can simplify the above result to obtain

$$X(\omega) = A e^{-j\omega t_0}.$$

Thus, we have shown that

$$A\delta(t - t_0) \xleftrightarrow{\mathcal{F}} A e^{-j\omega t_0}.$$

From the Fourier transform analysis and synthesis equations, we have that the Fourier transform representation of $x(t)$ is given by

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

where

$$X(\omega) = Ae^{-j\omega t_0}.$$

□

Example 5.2 (Inverse Fourier transform of the unit-impulse function). Find the inverse Fourier transform of $X(\omega) = 2\pi A\delta(\omega - \omega_0)$.

Solution. From the definition of the inverse Fourier transform, we can write

$$\begin{aligned}\mathcal{F}^{-1}\{2\pi A\delta(\omega - \omega_0)\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi A\delta(\omega - \omega_0)e^{j\omega t} d\omega \\ &= A \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t} d\omega.\end{aligned}$$

Using the sifting property of the unit-impulse function, we can simplify the preceding equation to obtain

$$\mathcal{F}^{-1}\{2\pi A\delta(\omega - \omega_0)\} = Ae^{j\omega_0 t}.$$

Thus, we have that

$$Ae^{j\omega_0 t} \xleftrightarrow{\mathcal{F}} 2\pi A\delta(\omega - \omega_0).$$

□

Example 5.3 (Fourier transform of the rectangular pulse). Find the Fourier transform $X(\omega)$ of the signal $x(t) = \text{rect}t$.

Solution. From the definition of the Fourier transform, we can write

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} [\text{rect}t]e^{-j\omega t} dt.$$

From the definition of the rectangular pulse function, we can simplify this equation as follows:

$$\begin{aligned}X(\omega) &= \int_{-1/2}^{1/2} [\text{rect}t]e^{-j\omega t} dt \\ &= \int_{-1/2}^{1/2} e^{-j\omega t} dt.\end{aligned}$$

Evaluating the integral and simplifying, we obtain

$$\begin{aligned}X(\omega) &= \left[-\frac{1}{j\omega} e^{-j\omega t} \right]_{-1/2}^{1/2} \\ &= \frac{1}{j\omega} [e^{j\omega/2} - e^{-j\omega/2}] \\ &= \frac{1}{j\omega} [2j \sin \frac{\omega}{2}] \\ &= \frac{2}{\omega} \sin \omega/2 \\ &= \text{sinc } \omega/2.\end{aligned}$$

Thus, we have shown that

$$\text{rect}t \xleftrightarrow{\mathcal{F}} \text{sinc } \omega/2.$$

□

5.4 Convergence of the Continuous-Time Fourier Transform

When deriving the Fourier transform representation earlier, we implicitly made some assumptions about the convergence of the integrals and other expressions involved. These assumptions are not always valid. For this reason, a more careful examination of the convergence properties of the Fourier transform is in order.

Suppose that we have an arbitrary signal $x(t)$. This signal has the Fourier transform representation $\hat{x}(t)$ given by

$$\hat{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega,$$

where

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt.$$

Now, we need to concern ourselves with the convergence properties of this representation. In other words, we want to know when $\hat{x}(t)$ is a valid representation of $x(t)$. In our earlier derivation of the Fourier transform, we relied heavily on the Fourier series. Therefore, one might expect that the convergence of the Fourier transform representation is closely related to the convergence properties of Fourier series. This is, in fact, the case. The convergence properties of the Fourier transform are very similar to the convergence properties of the Fourier series (as studied in Section 4.4).

The first important result concerning convergence relates to finite-energy signals as stated by the theorem below.

Theorem 5.1 (Convergence of Fourier transform (finite-energy case)). *If a signal $x(t)$ is of finite energy (i.e., $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$), then its Fourier transform representation converges in the MSE sense.*

In other words, if $x(t)$ is of finite energy, then

$$E = \int_{-\infty}^{\infty} |\hat{x}(t) - x(t)|^2 dt = 0.$$

Although $x(t)$ and $\hat{x}(t)$ may differ at individual points, the energy E in the difference is zero.

The other important result concerning convergence that we shall consider relates to what are known as the Dirichlet conditions. The Dirichlet conditions for the signal $x(t)$ are as follows:

1. The signal $x(t)$ is absolutely integrable (i.e., $\int_{-\infty}^{\infty} |x(t)| dt < \infty$).
2. The signal $x(t)$ has a finite number of maxima and minima on any finite interval.
3. The signal $x(t)$ has a finite number of discontinuities on any finite interval, and each discontinuity is itself finite.

For a signal satisfying the Dirichlet conditions, we have the important convergence result stated below.

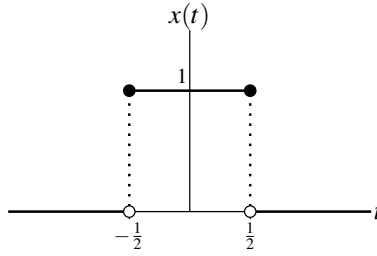
Theorem 5.2 (Convergence of Fourier transform (Dirichlet case)). *If a signal $x(t)$ satisfies the Dirichlet conditions, then its Fourier transform representation $\hat{x}(t)$ converges pointwise for all t , except at points of discontinuity. Furthermore, at each discontinuity point $t = t_a$, we have that*

$$\hat{x}(t_a) = \frac{1}{2} [x(t_a^+) + x(t_a^-)],$$

where $x(t_a^-)$ and $x(t_a^+)$ denote the values of the signal $x(t)$ on the left- and right-hand sides of the discontinuity, respectively.

In other words, if a signal $x(t)$ satisfies the Dirichlet conditions, then the Fourier transform representation $\hat{x}(t)$ of $x(t)$ converges to $x(t)$ for all t , except at points of discontinuity where $\hat{x}(t)$ instead converges to the average of $x(t)$ on the two sides of the discontinuity.

The finite-energy and Dirichlet conditions mentioned above are only sufficient conditions for the convergence of the Fourier transform representation. They are not necessary. In other words, a signal may violate these conditions and still have a valid Fourier transform representation.

Figure 5.4: Signal $x(t)$.

Example 5.4. Consider the function $x(t)$ shown in Figure 5.4. Let $\hat{x}(t)$ denote the Fourier transform representation of $x(t)$ (i.e., $\hat{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$, where $X(\omega)$ denotes the Fourier transform of $x(t)$). Determine the values $\hat{x}(-\frac{1}{2})$ and $\hat{x}(\frac{1}{2})$.

Solution. We begin by observing that $x(t)$ satisfies the Dirichlet conditions. Consequently, Theorem 5.2 applies. Thus, we have that

$$\begin{aligned}\hat{x}\left(-\frac{1}{2}\right) &= \frac{1}{2} \left[x\left(-\frac{1}{2}^{-}\right) + x\left(-\frac{1}{2}^{+}\right) \right] = \frac{1}{2}(0 + 1) = \frac{1}{2} \quad \text{and} \\ \hat{x}\left(\frac{1}{2}\right) &= \frac{1}{2} \left[x\left(\frac{1}{2}^{-}\right) + x\left(\frac{1}{2}^{+}\right) \right] = \frac{1}{2}(1 + 0) = \frac{1}{2}.\end{aligned}$$

□

5.5 Properties of the Continuous-Time Fourier Transform

The Fourier transform has a number of important properties. In the sections that follow, we introduce several of these properties. For convenience, these properties are also later summarized in Table 5.1 (on page 123).

5.5.1 Linearity

If $x_1(t) \xleftrightarrow{\mathcal{F}} X_1(\omega)$ and $x_2(t) \xleftrightarrow{\mathcal{F}} X_2(\omega)$, then

$$a_1 x_1(t) + a_2 x_2(t) \xleftrightarrow{\mathcal{F}} a_1 X_1(\omega) + a_2 X_2(\omega),$$

where a_1 and a_2 are arbitrary complex constants. This is known as the linearity property of the Fourier transform.

To prove the above property, we proceed as follows:

$$\begin{aligned}\mathcal{F}\{a_1 x_1(t) + a_2 x_2(t)\} &= \int_{-\infty}^{\infty} [a_1 x_1(t) + a_2 x_2(t)] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} a_1 x_1(t) e^{-j\omega t} dt + \int_{-\infty}^{\infty} a_2 x_2(t) e^{-j\omega t} dt \\ &= a_1 \int_{-\infty}^{\infty} x_1(t) e^{-j\omega t} dt + a_2 \int_{-\infty}^{\infty} x_2(t) e^{-j\omega t} dt \\ &= a_1 \mathcal{F}\{x_1(t)\} + a_2 \mathcal{F}\{x_2(t)\} \\ &= a_1 X_1(\omega) + a_2 X_2(\omega).\end{aligned}$$

Thus, we have shown that the linearity property holds.

Example 5.5 (Linearity property of the Fourier transform). Find the Fourier transform $X(\omega)$ of the signal $x(t) = A \cos \omega_0 t$.

Solution. We recall that $\cos \alpha = \frac{1}{2}[e^{j\alpha} + e^{-j\alpha}]$ for any real α . Thus, we can write

$$\begin{aligned} X(\omega) &= \mathcal{F}\{x(t)\} \\ &= \mathcal{F}\{A \cos \omega_0 t\} \\ &= \mathcal{F}\left\{\frac{A}{2}[e^{j\omega_0 t} + e^{-j\omega_0 t}]\right\}. \end{aligned}$$

Then, we use the linearity property of the Fourier transform to obtain

$$X(\omega) = \frac{A}{2} \mathcal{F}\{e^{j\omega_0 t}\} + \frac{A}{2} \mathcal{F}\{e^{-j\omega_0 t}\}.$$

From Example 5.2, we know that $e^{j\omega_0 t} \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega - \omega_0)$ and $e^{-j\omega_0 t} \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega + \omega_0)$. Thus, we can further simplify the above expression for $X(\omega)$ as follows:

$$\begin{aligned} X(\omega) &= \frac{A}{2}[2\pi\delta(\omega + \omega_0)] + \frac{A}{2}[2\pi\delta(\omega - \omega_0)] \\ &= A\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]. \end{aligned}$$

Thus, we have shown that

$$A \cos \omega_0 t \xleftrightarrow{\mathcal{F}} A\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)].$$

□

Example 5.6 (Fourier transform of the unit-step function). Suppose that $\text{sgn } t \xleftrightarrow{\mathcal{F}} \frac{2}{j\omega}$. Find the Fourier transform $X(\omega)$ of the signal $x(t) = u(t)$.

Solution. First, we observe that $x(t)$ can be expressed in terms of the signum function as

$$x(t) = u(t) = \frac{1}{2} + \frac{1}{2} \text{sgn } t.$$

Taking the Fourier transform of both sides of this equation yields

$$X(\omega) = \mathcal{F}\{x(t)\} = \mathcal{F}\left\{\frac{1}{2} + \frac{1}{2} \text{sgn } t\right\}.$$

Using the linearity property of the Fourier transform, we can rewrite this as

$$X(\omega) = \frac{1}{2} \mathcal{F}\{1\} + \frac{1}{2} \mathcal{F}\{\text{sgn } t\}.$$

From Example 5.5 (with $\omega_0 = 0$), we know that $1 \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega)$. Also, we are given that $\text{sgn } t \xleftrightarrow{\mathcal{F}} \frac{2}{j\omega}$. Using these facts, we can rewrite the expression for $X(\omega)$ as

$$\begin{aligned} X(\omega) &= \frac{1}{2}[2\pi\delta(\omega)] + \frac{1}{2}\left(\frac{2}{j\omega}\right) \\ &= \pi\delta(\omega) + \frac{1}{j\omega}. \end{aligned}$$

Thus, we have shown that

$$u(t) \xleftrightarrow{\mathcal{F}} \pi\delta(\omega) + \frac{1}{j\omega}.$$

□

5.5.2 Time-Domain Shifting

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j\omega t_0} X(\omega),$$

where t_0 is an arbitrary real constant. This is known as the time-shifting property of the Fourier transform.

To prove the above property, we proceed as follows. To begin, we have

$$\mathcal{F}\{x(t - t_0)\} = \int_{-\infty}^{\infty} x(t - t_0) e^{-j\omega t} dt.$$

Now, we use a change of variable. Let $\lambda = t - t_0$ so that $t = \lambda + t_0$ and $dt = d\lambda$. Performing the change of variable and simplifying, we obtain

$$\begin{aligned} \mathcal{F}\{x(t - t_0)\} &= \int_{-\infty}^{\infty} x(\lambda) e^{-j\omega(\lambda + t_0)} d\lambda \\ &= \int_{-\infty}^{\infty} x(\lambda) e^{-j\omega\lambda} e^{-j\omega t_0} d\lambda \\ &= e^{-j\omega t_0} \int_{-\infty}^{\infty} x(\lambda) e^{-j\omega\lambda} d\lambda \\ &= e^{-j\omega t_0} \mathcal{F}\{x(t)\} \\ &= e^{-j\omega t_0} X(\omega). \end{aligned}$$

Thus, we have proven that the time-shifting property holds.

Example 5.7 (Time-domain shifting property of the Fourier transform). Find the Fourier transform $X(\omega)$ of the signal $x(t) = A \cos(\omega_0 t + \theta)$.

Solution. Let $v(t) = A \cos \omega_0 t$ so that $x(t) = v(t + \frac{\theta}{\omega_0})$. From Example 5.5, we have that

$$\begin{aligned} \mathcal{F}\{v(t)\} &= V(\omega) = \mathcal{F}\{A \cos \omega_0 t\} \\ &= A\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]. \end{aligned}$$

From the definition of $v(t)$ and the time-shifting property of the Fourier transform, we have

$$\begin{aligned} X(\omega) &= \mathcal{F}\{x(t)\} \\ &= e^{j\omega\theta/\omega_0} V(\omega) \\ &= e^{j\omega\theta/\omega_0} A\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]. \end{aligned}$$

Thus, we have shown that

$$A \cos(\omega_0 t + \theta) \xleftrightarrow{\mathcal{F}} A\pi e^{j\omega\theta/\omega_0} [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)].$$

□

5.5.3 Frequency-Domain Shifting

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$e^{j\omega_0 t} x(t) \xleftrightarrow{\mathcal{F}} X(\omega - \omega_0),$$

where ω_0 is an arbitrary real constant. This is known as the frequency-domain shifting property of the Fourier transform.

To prove the above property, we proceed as follows. From the definition of the Fourier transform and straightforward algebraic manipulation, we can write

$$\begin{aligned}\mathcal{F}\{e^{j\omega_0 t}x(t)\} &= \int_{-\infty}^{\infty} e^{j\omega_0 t}x(t)e^{-j\omega t}dt \\ &= \int_{-\infty}^{\infty} x(t)e^{-j(\omega-\omega_0)t}dt \\ &= X(\omega-\omega_0).\end{aligned}$$

Thus, we have shown that the frequency-domain shifting property holds.

Example 5.8 (Frequency-domain shifting property of the Fourier transform). Find the Fourier transform $X(\omega)$ of the signal $x(t) = (\cos \omega_0 t)(\cos 20\pi t)$.

Solution. Recall that $\cos \alpha = \frac{1}{2}[e^{j\alpha} + e^{-j\alpha}]$ for any real α . Using this relationship and the linearity property of the Fourier transform, we can write

$$\begin{aligned}X(\omega) &= \mathcal{F}\{x(t)\} \\ &= \mathcal{F}\{(\cos \omega_0 t)\left(\frac{1}{2}\right)[e^{j20\pi t} + e^{-j20\pi t}]\} \\ &= \mathcal{F}\left\{\frac{1}{2}e^{j20\pi t}\cos \omega_0 t + \frac{1}{2}e^{-j20\pi t}\cos \omega_0 t\right\} \\ &= \frac{1}{2}\mathcal{F}\{e^{j20\pi t}\cos \omega_0 t\} + \frac{1}{2}\mathcal{F}\{e^{-j20\pi t}\cos \omega_0 t\}.\end{aligned}$$

From Example 5.5 (where we showed $\cos \omega_0 t \xleftrightarrow{\mathcal{F}} \pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$) and the frequency-domain shifting property of the Fourier transform, we have

$$\begin{aligned}X(\omega) &= \frac{1}{2}[\pi[\delta(\nu-\omega_0) + \delta(\nu+\omega_0)]]|_{\nu=\omega-20\pi} + \frac{1}{2}[\pi[\delta(\nu-\omega_0) + \delta(\nu+\omega_0)]]|_{\nu=\omega+20\pi} \\ &= \frac{1}{2}(\pi[\delta(\omega+\omega_0-20\pi) + \delta(\omega-\omega_0-20\pi)] + \pi[\delta(\omega+\omega_0+20\pi) + \delta(\omega-\omega_0+20\pi)]) \\ &= \frac{\pi}{2}[\delta(\omega+\omega_0-20\pi) + \delta(\omega-\omega_0-20\pi) + \delta(\omega+\omega_0+20\pi) + \delta(\omega-\omega_0+20\pi)].\end{aligned}$$

□

5.5.4 Time- and Frequency-Domain Scaling

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|}X\left(\frac{\omega}{a}\right),$$

where a is an arbitrary nonzero real constant. This is known as the time/frequency-scaling property of the Fourier transform.

To prove the above property, we proceed as follows. From the definition of the Fourier transform, we can write

$$\mathcal{F}\{x(at)\} = \int_{-\infty}^{\infty} x(at)e^{-j\omega t}dt.$$

Now, we use a change of variable. Let $\lambda = at$ so that $t = \lambda/a$ and $dt = d\lambda/a$. Performing the change of variable (and being mindful of the change in the limits of integration), we obtain

$$\begin{aligned}\mathcal{F}\{x(at)\} &= \begin{cases} \int_{-\infty}^{\infty} x(\lambda)e^{-j(\omega/a)\lambda}\left(\frac{1}{a}\right)d\lambda & \text{for } a > 0 \\ \int_{\infty}^{-\infty} x(\lambda)e^{-j(\omega/a)\lambda}\left(\frac{1}{a}\right)d\lambda & \text{for } a < 0 \end{cases} \\ &= \begin{cases} \frac{1}{a} \int_{-\infty}^{\infty} x(\lambda)e^{-j(\omega/a)\lambda}d\lambda & \text{for } a > 0 \\ -\frac{1}{a} \int_{-\infty}^{\infty} x(\lambda)e^{-j(\omega/a)\lambda}d\lambda & \text{for } a < 0. \end{cases}\end{aligned}$$

Combining the two cases (i.e., for $a < 0$ and $a > 0$), we obtain

$$\begin{aligned}\mathcal{F}\{x(at)\} &= \frac{1}{|a|} \int_{-\infty}^{\infty} x(\lambda) e^{-j(\omega/a)\lambda} d\lambda \\ &= \frac{1}{|a|} X\left(\frac{\omega}{a}\right).\end{aligned}$$

Thus, we have shown that the time/frequency-scaling property holds.

Example 5.9 (Time scaling property of the Fourier transform). Find the Fourier transform $X(\omega)$ of the signal $x(t) = \text{rect}(at)$.

Solution. Let $v(t) = \text{rect}(t)$ so that $x(t) = v(at)$. From Example 5.3, we know that

$$\mathcal{F}\{v(t)\} = V(\omega) = \mathcal{F}\{\text{rect}t\} = \text{sinc } \omega/2. \quad (5.10)$$

From the definition of $v(t)$ and the time-scaling property of the Fourier transform, we have

$$\begin{aligned}X(\omega) &= \mathcal{F}\{x(t)\} \\ &= \frac{1}{|a|} V\left(\frac{\omega}{a}\right).\end{aligned}$$

Substituting the expression for $V(\omega)$ in (5.10) into the preceding equation, we have

$$X(\omega) = \frac{1}{|a|} \text{sinc } \frac{\omega}{2a}.$$

Thus, we have shown that

$$\text{rect}(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} \text{sinc } \frac{\omega}{2a}.$$

□

5.5.5 Conjugation

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$x^*(t) \xleftrightarrow{\mathcal{F}} X^*(-\omega).$$

This is known as the conjugation property of the Fourier transform.

A proof of the above property is quite simple. From the definition of the Fourier transform, we have

$$\mathcal{F}\{x^*(t)\} = \int_{-\infty}^{\infty} x^*(t) e^{-j\omega t} dt.$$

From the properties of conjugation, we can rewrite this equation as

$$\begin{aligned}\mathcal{F}\{x^*(t)\} &= \left[\left(\int_{-\infty}^{\infty} x^*(t) e^{-j\omega t} dt \right)^* \right]^* \\ &= \left[\int_{-\infty}^{\infty} [x(t)]^* [e^{-j\omega t}]^* dt \right]^* \\ &= \left[\int_{-\infty}^{\infty} x(t) e^{j\omega t} dt \right]^* \\ &= X^*(-\omega).\end{aligned}$$

Thus, we have shown that the conjugation property holds.

Example 5.10 (Fourier transform of a real signal). Show that the Fourier transform $X(\omega)$ of any real signal $x(t)$ must satisfy $X(\omega) = X^*(-\omega)$, and this condition implies that $|X(\omega)| = |X(-\omega)|$ and $\arg X(\omega) = -\arg X(-\omega)$ (i.e., $|X(\omega)|$ and $\arg X(\omega)$ are even and odd functions, respectively).

Solution. From the conjugation property of the Fourier transform, we have

$$\mathcal{F}\{x^*(t)\} = X^*(-\omega).$$

Since $x(t)$ is real, $x(t) = x^*(t)$ which implies

$$\mathcal{F}\{x(t)\} = X^*(-\omega),$$

or equivalently

$$X(\omega) = X^*(-\omega). \quad (5.11)$$

Taking the magnitude of both sides of (5.11) and observing that $|z| = |z^*|$ for any complex z , we have

$$\begin{aligned} |X(\omega)| &= |X^*(-\omega)| \\ &= |X(-\omega)|. \end{aligned}$$

Taking the argument of both sides of (5.11) and observing that $\arg z^* = -\arg z$ for any complex z , we have

$$\begin{aligned} \arg X(\omega) &= \arg X^*(-\omega) \\ &= -\arg X(-\omega). \end{aligned}$$

□

5.5.6 Duality

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$X(t) \xleftrightarrow{\mathcal{F}} 2\pi x(-\omega)$$

This is known as the duality property of the Fourier transform. This property follows from the similarity/symmetry in the definition of the forward and inverse Fourier transforms. That is, the forward Fourier transform equation given by (5.8) and the inverse Fourier transform equation given by (5.9) are identical except for a factor of 2π and different sign in the parameter for the exponential function.

To prove the above property, we proceed as follows. From the Fourier transform synthesis equation, we have

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega.$$

Substituting $-t$ for t , we obtain

$$x(-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega.$$

Now, we multiply both sides of the equation by 2π to yield

$$\begin{aligned} 2\pi x(-t) &= \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega \\ &= \mathcal{F}\{X(t)\}. \end{aligned}$$

Thus, we have shown that the duality property holds.

Example 5.11 (Fourier transform of the sinc function). Given that $\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc } \omega/2$, find the Fourier transform $X(\omega)$ of the signal $x(t) = \text{sinc}(t/2)$.

Solution. From the given Fourier transform pair and the duality property, we have that

$$X(\omega) = \mathcal{F}\{\text{sinc } t/2\} = 2\pi \text{rect}(-\omega).$$

Since $\text{rect}(-\omega) = \text{rect}(\omega)$, we can simplify this to obtain

$$X(\omega) = 2\pi \text{rect}(\omega).$$

Thus, we have shown that

$$\text{sinc } t/2 \xleftrightarrow{\mathcal{F}} 2\pi \text{rect } \omega.$$

□

5.5.7 Time-Domain Convolution

If $x_1(t) \xleftrightarrow{\mathcal{F}} X_1(\omega)$ and $x_2(t) \xleftrightarrow{\mathcal{F}} X_2(\omega)$, then

$$x_1(t) * x_2(t) \xleftrightarrow{\mathcal{F}} X_1(\omega)X_2(\omega).$$

This is known as the time-domain convolution property of the Fourier transform.

The proof of this property is as follows. From the definition of the Fourier transform and convolution, we have

$$\begin{aligned} \mathcal{F}\{x_1(t) * x_2(t)\} &= \int_{-\infty}^{\infty} [x_1(t) * x_2(t)] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) d\tau \right] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) e^{-j\omega t} d\tau dt. \end{aligned}$$

Changing the order of integration, we obtain

$$\mathcal{F}\{x_1(t) * x_2(t)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) e^{-j\omega t} dt d\tau.$$

Now, we use a change of variable. Let $\lambda = t - \tau$ so that $t = \lambda + \tau$ and $d\lambda = dt$. Applying the change of variable and simplifying, we obtain

$$\begin{aligned} \mathcal{F}\{x_1(t) * x_2(t)\} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1(\tau) x_2(\lambda) e^{-j\omega(\lambda + \tau)} d\lambda d\tau \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1(\tau) x_2(\lambda) e^{-j\omega\lambda} e^{-j\omega\tau} d\lambda d\tau \\ &= \int_{-\infty}^{\infty} x_1(\tau) e^{-j\omega\tau} \left[\int_{-\infty}^{\infty} x_2(\lambda) e^{-j\omega\lambda} d\lambda \right] d\tau \\ &= \left[\int_{-\infty}^{\infty} x_1(\tau) e^{-j\omega\tau} d\tau \right] \left[\int_{-\infty}^{\infty} x_2(\lambda) e^{-j\omega\lambda} d\lambda \right] \\ &= \mathcal{F}\{x_1(t)\} \mathcal{F}\{x_2(t)\} \\ &= X_1(\omega) X_2(\omega). \end{aligned}$$

Thus, we have shown that the time-domain convolution property holds.

Example 5.12 (Time-domain convolution property of the Fourier transform). With the aid of Table 5.2, find the Fourier transform $X(\omega)$ of the signal $x(t) = x_1(t) * x_2(t)$ where

$$\begin{aligned} x_1(t) &= e^{-2t}u(t) \quad \text{and} \\ x_2(t) &= u(t). \end{aligned}$$

Solution. Let $X_1(\omega)$ and $X_2(\omega)$ denote the Fourier transforms of $x_1(t)$ and $x_2(t)$, respectively. From the time-domain convolution property of the Fourier transform, we know that

$$X(\omega) = \mathcal{F}\{x_1(t) * x_2(t)\} = X_1(\omega)X_2(\omega). \quad (5.12)$$

From Table 5.2, we know that

$$\begin{aligned} X_1(\omega) &= \mathcal{F}\{x_1(t)\} = \mathcal{F}\{e^{-2t}u(t)\} = \frac{1}{2+j\omega} \quad \text{and} \\ X_2(\omega) &= \mathcal{F}\{x_2(t)\} = \mathcal{F}\{u(t)\} = \pi\delta(\omega) + \frac{1}{j\omega}. \end{aligned}$$

Substituting these expressions for $X_1(\omega)$ and $X_2(\omega)$ into (5.12), we obtain

$$\begin{aligned} X(\omega) &= \left[\frac{1}{2+j\omega}\right]\left(\pi\delta(\omega) + \frac{1}{j\omega}\right) \\ &= \frac{\pi}{2+j\omega}\delta(\omega) + \frac{1}{j\omega}\left(\frac{1}{2+j\omega}\right) \\ &= \frac{\pi}{2+j\omega}\delta(\omega) + \frac{1}{j2\omega-\omega^2} \\ &= \frac{\pi}{2}\delta(\omega) + \frac{1}{j2\omega-\omega^2}. \end{aligned}$$

□

5.5.8 Frequency-Domain Convolution

If $x_1(t) \xleftrightarrow{\mathcal{F}} X_1(\omega)$ and $x_2(t) \xleftrightarrow{\mathcal{F}} X_2(\omega)$, then

$$x_1(t)x_2(t) \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi}X_1(\omega) * X_2(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\theta)X_2(\omega - \theta)d\theta.$$

This is known as the frequency-domain convolution (or time-domain multiplication) property of the Fourier transform.

To prove the above property, we proceed as follows. From the definition of the inverse Fourier transform, we have

$$\begin{aligned} \mathcal{F}^{-1}\left\{\frac{1}{2\pi}X_1(\omega) * X_2(\omega)\right\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{1}{2\pi}X_1(\omega) * X_2(\omega)\right] e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \frac{1}{2\pi}X_1(\lambda)X_2(\omega - \lambda)d\lambda\right] e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi}X_1(\lambda)X_2(\omega - \lambda)e^{j\omega t} d\lambda d\omega. \end{aligned}$$

Reversing the order of integration, we obtain

$$\mathcal{F}^{-1}\left\{\frac{1}{2\pi}X_1(\omega) * X_2(\omega)\right\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi}X_1(\lambda)X_2(\omega - \lambda)e^{j\omega t} d\omega d\lambda.$$

Now, we employ a change of variable. Let $v = \omega - \lambda$ so that $\omega = v + \lambda$ and $dv = d\lambda$. Applying the change of variable

and simplifying yields

$$\begin{aligned}
 \mathcal{F}^{-1} \left\{ \frac{1}{2\pi} X_1(\omega) * X_2(\omega) \right\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} X_1(\lambda) X_2(v) e^{j(v+\lambda)t} dv d\lambda \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} X_1(\lambda) X_2(v) e^{jv t} e^{j\lambda t} dv d\lambda \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\lambda) e^{j\lambda t} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X_2(v) e^{jv t} dv \right] d\lambda \\
 &= \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\lambda) e^{j\lambda t} d\lambda \right] \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X_2(v) e^{jv t} dv \right] \\
 &= \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) e^{j\omega t} d\omega \right] \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X_2(\omega) e^{j\omega t} d\omega \right] \\
 &= \mathcal{F}^{-1} \{X_1(\omega)\} \mathcal{F}^{-1} \{X_2(\omega)\} \\
 &= x_1(t) x_2(t).
 \end{aligned}$$

Thus, we have shown that the frequency-domain convolution property holds.

Example 5.13 (Frequency-domain convolution property). Suppose that we have the signal

$$y(t) = x(t) \cos \omega_c t$$

where ω_c is a nonzero real constant. Find the Fourier transform $Y(\omega)$ of the signal $y(t)$ in terms of $X(\omega) = \mathcal{F}\{x(t)\}$.

Solution. Taking the Fourier transform of both sides of the above equation for $y(t)$, we have

$$Y(\omega) = \mathcal{F}\{x(t) \cos \omega_c t\}.$$

Using the frequency-domain convolution property of the Fourier transform, we can write

$$\begin{aligned}
 Y(\omega) &= \mathcal{F}\{x(t) \cos \omega_c t\} \\
 &= \frac{1}{2\pi} \mathcal{F}\{x(t)\} * \mathcal{F}\{\cos \omega_c t\} \\
 &= \frac{1}{2\pi} X(\omega) * [\pi[\delta(\omega - \omega_c) + \delta(\omega + \omega_c)]] \\
 &= \frac{1}{2} X(\omega) * [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] \\
 &= \frac{1}{2} [X(\omega) * \delta(\omega - \omega_c) + X(\omega) * \delta(\omega + \omega_c)] \\
 &= \frac{1}{2} \left[\int_{-\infty}^{\infty} X(\lambda) \delta(\omega - \lambda - \omega_c) d\lambda + \int_{-\infty}^{\infty} X(\lambda) \delta(\omega - \lambda + \omega_c) d\lambda \right] \\
 &= \frac{1}{2} \left[\int_{-\infty}^{\infty} X(\lambda) \delta(\lambda - \omega + \omega_c) d\lambda + \int_{-\infty}^{\infty} X(\lambda) \delta(\lambda - \omega - \omega_c) d\lambda \right] \\
 &= \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)] \\
 &= \frac{1}{2} X(\omega - \omega_c) + \frac{1}{2} X(\omega + \omega_c).
 \end{aligned}$$

□

5.5.9 Time-Domain Differentiation

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$\frac{dx(t)}{dt} \xleftrightarrow{\mathcal{F}} j\omega X(\omega).$$

This is known as the time-domain differentiation property of the Fourier transform.

To prove the above property, we begin by using the definition of the inverse Fourier transform to write

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega.$$

Now, we differentiate both sides of the preceding equation with respect to t and simplify to obtain

$$\begin{aligned} \frac{dx(t)}{dt} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) (j\omega) e^{j\omega t} d\omega \\ &= \mathcal{F}^{-1}\{j\omega X(\omega)\}. \end{aligned}$$

Thus, we have shown that the time-differentiation property holds. In passing, we also note that by repeating the above argument, we have

$$\frac{d^n}{dt^n} x(t) \xleftrightarrow{\mathcal{F}} (j\omega)^n X(\omega).$$

Example 5.14 (Time-domain differentiation property). Find the Fourier transform $X(\omega)$ of the signal $x(t) = \frac{d}{dt} \delta(t)$.

Solution. Taking the Fourier transform of both sides of the above equation for $x(t)$ yields

$$X(\omega) = \mathcal{F}\left\{\frac{d}{dt} \delta(t)\right\}.$$

Using the time-domain differentiation property of the Fourier transform, we can write

$$\begin{aligned} X(\omega) &= \mathcal{F}\left\{\frac{d}{dt} \delta(t)\right\} \\ &= j\omega \mathcal{F}\{\delta(t)\} \\ &= j\omega(1) \\ &= j\omega. \end{aligned}$$

□

5.5.10 Frequency-Domain Differentiation

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$tx(t) \xleftrightarrow{\mathcal{F}} j \frac{d}{d\omega} X(\omega).$$

This is known as the frequency-domain differentiation property of the Fourier transform.

To prove the above property, we proceed as follows. From the definition of the Fourier transform, we can write

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt.$$

Now, we differentiate both sides of this equation with respect to ω and simplify to obtain

$$\begin{aligned} \frac{d}{d\omega} X(\omega) &= \int_{-\infty}^{\infty} x(t) (-jt) e^{-j\omega t} dt \\ &= -j \int_{-\infty}^{\infty} tx(t) e^{-j\omega t} dt \\ &= -j \mathcal{F}\{tx(t)\}. \end{aligned}$$

Multiplying both sides of the preceding equation by j yields

$$j \frac{d}{d\omega} X(\omega) = \mathcal{F}\{tx(t)\}.$$

Thus, we have shown that the frequency-domain differentiation property holds.

Example 5.15 (Frequency-domain differentiation property). Find the Fourier transform $X(\omega)$ of the signal $x(t) = t \cos \omega_0 t$ where ω_0 is a nonzero real constant.

Solution. Taking the Fourier transform of both sides of the equation for $x(t)$ yields

$$X(\omega) = \mathcal{F}\{t \cos \omega_0 t\}.$$

From the frequency-domain differentiation property of the Fourier transform, we can write

$$\begin{aligned} X(\omega) &= \mathcal{F}\{t \cos \omega_0 t\} \\ &= j \frac{d}{d\omega} [\mathcal{F}\{\cos \omega_0 t\}] \\ &= j \frac{d}{d\omega} [\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]] \\ &= j\pi \frac{d}{d\omega} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \\ &= j\pi \frac{d}{d\omega} \delta(\omega - \omega_0) + j\pi \frac{d}{d\omega} \delta(\omega + \omega_0). \end{aligned}$$

□

5.5.11 Time-Domain Integration

If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega} X(\omega) + \pi X(0) \delta(\omega).$$

This is known as the time-domain integration property of the Fourier transform.

The above property can be proven as follows. First, we observe that

$$\int_{-\infty}^t x(\tau) d\tau = x(t) * u(t).$$

Taking the Fourier transform of both sides of the preceding equation and using the time-domain convolution property of the Fourier transform, we have

$$\begin{aligned} \mathcal{F}\left\{\int_{-\infty}^t x(\tau) d\tau\right\} &= \mathcal{F}\{x(t) * u(t)\} \\ &= X(\omega) \mathcal{F}\{u(t)\}. \end{aligned} \tag{5.13}$$

From Example 5.6, we know that $u(t) \xleftrightarrow{\mathcal{F}} \pi \delta(\omega) + \frac{1}{j\omega}$. Using this fact, we can rewrite (5.13) as

$$\begin{aligned} \mathcal{F}\left\{\int_{-\infty}^t x(\tau) d\tau\right\} &= X(\omega) [\pi \delta(\omega) + \frac{1}{j\omega}] \\ &= \frac{1}{j\omega} X(\omega) + \pi X(0) \delta(\omega). \end{aligned}$$

From the equivalence property of the unit-impulse function, we have

$$\mathcal{F}\left\{\int_{-\infty}^t x(\tau) d\tau\right\} = \frac{1}{j\omega} X(\omega) + \pi X(0) \delta(\omega).$$

Thus, we have shown that the time-domain integration property holds.

Example 5.16 (Time-domain integration property of the Fourier transform). Use the time-domain integration property of the Fourier transform in order to find the Fourier transform $X(\omega)$ of the signal $x(t) = u(t)$.

Solution. We begin by observing that $x(t)$ can be expressed in terms of an integral as follows:

$$x(t) = u(t) = \int_{-\infty}^t \delta(\tau) d\tau.$$

Now, we consider the Fourier transform of $x(t)$. We have

$$\begin{aligned} X(\omega) &= \mathcal{F}\{u(t)\} \\ &= \mathcal{F}\left\{\int_{-\infty}^t \delta(\tau) d\tau\right\}. \end{aligned}$$

From the time-domain integration property, we can write

$$\begin{aligned} X(\omega) &= \frac{1}{j\omega} \mathcal{F}\{\delta(t)\} + \pi [\mathcal{F}\{\delta(t)\}]|_{\omega=0} \delta(\omega) \\ &= \frac{1}{j\omega} (1) + \pi(1) \delta(\omega) \\ &= \frac{1}{j\omega} + \pi \delta(\omega). \end{aligned}$$

Thus, we have shown that $u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega} + \pi \delta(\omega)$. □

5.5.12 Parseval's Relation

Recall that the energy of a signal $f(t)$ is given by $\int_{-\infty}^{\infty} |f(t)|^2 dt$. If $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$, then

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega. \quad (5.14)$$

That is, the energy of $x(t)$ and energy of $X(\omega)$ are equal within a scaling factor of 2π . This relationship is known as Parseval's relation.

To prove the above relationship, we proceed as follows. Consider the left-hand side of (5.14) which we can write as

$$\begin{aligned} \int_{-\infty}^{\infty} |x(t)|^2 dt &= \int_{-\infty}^{\infty} x(t) x^*(t) dt \\ &= \int_{-\infty}^{\infty} x(t) \mathcal{F}^{-1}\{\mathcal{F}\{x^*(t)\}\} dt. \end{aligned}$$

From the conjugation property of the Fourier transform, we have that $x^*(t) \xleftrightarrow{\mathcal{F}} X^*(-\omega)$. So, we can rewrite the above equation as

$$\begin{aligned} \int_{-\infty}^{\infty} |x(t)|^2 dt &= \int_{-\infty}^{\infty} x(t) \mathcal{F}^{-1}\{X^*(-\omega)\} dt \\ &= \int_{-\infty}^{\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(-\omega) e^{j\omega t} d\omega \right] dt. \end{aligned}$$

Now, we employ a change of variable (i.e., replace ω by $-\omega$) to obtain

$$\begin{aligned} \int_{-\infty}^{\infty} |x(t)|^2 dt &= \int_{-\infty}^{\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega \right] dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) X^*(\omega) e^{-j\omega t} d\omega dt. \end{aligned}$$

Reversing the order of integration and simplifying, we have

$$\begin{aligned}
 \int_{-\infty}^{\infty} |x(t)|^2 dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) X^*(\omega) e^{-j\omega t} dt d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) \left[\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right] d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) X(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega.
 \end{aligned}$$

Thus, Parseval's relation holds.

Example 5.17 (Energy of the sinc signal). Suppose that we have the signal $x(t) = \text{sinc } t/2$ and $\mathcal{F}\{x(t)\} = 2\pi \text{rect } \omega$. Compute the energy of the signal $x(t)$.

Solution. We could directly compute the energy of the signal $x(t)$ as

$$\begin{aligned}
 E &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\
 &= \int_{-\infty}^{\infty} |\text{sinc } t/2|^2 dt.
 \end{aligned}$$

This integral is not so easy to compute, however. Instead, we use Parseval's relation to write

$$\begin{aligned}
 E &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |2\pi \text{rect } \omega|^2 d\omega \\
 &= \frac{1}{2\pi} \int_{-1/2}^{1/2} (2\pi)^2 d\omega \\
 &= 2\pi \int_{-1/2}^{1/2} d\omega \\
 &= 2\pi [\omega]_{-1/2}^{1/2} \\
 &= 2\pi \left[\frac{1}{2} + \frac{1}{2} \right] \\
 &= 2\pi.
 \end{aligned}$$

Thus, we have

$$E = \int_{-\infty}^{\infty} \left| \text{sinc } \frac{t}{2} \right|^2 dt = 2\pi.$$

□

5.6 Continuous-Time Fourier Transform of Periodic Signals

The Fourier transform can be applied to both aperiodic and periodic signals. In what follows, we derive a method for computing the Fourier transform of a periodic signal.

Suppose that we have a periodic signal $x(t)$ with period T . Since $x(t)$ is periodic, we can express it using a Fourier series as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad (5.15)$$